

MLX Cleanup Benchmark

Replacing a 3.3 GB Ollama cleanup hop with a lightweight native MLX model

Research brief | Run label: task-2-formal-cleanup-benchmark | July 6, 2026 | Local benchmark over 24 filler-heavy dictation references

0.240s

median cleanup latency
MLX after unloading 30B
model

3.8x

median speedup
0.914s / 0.240s

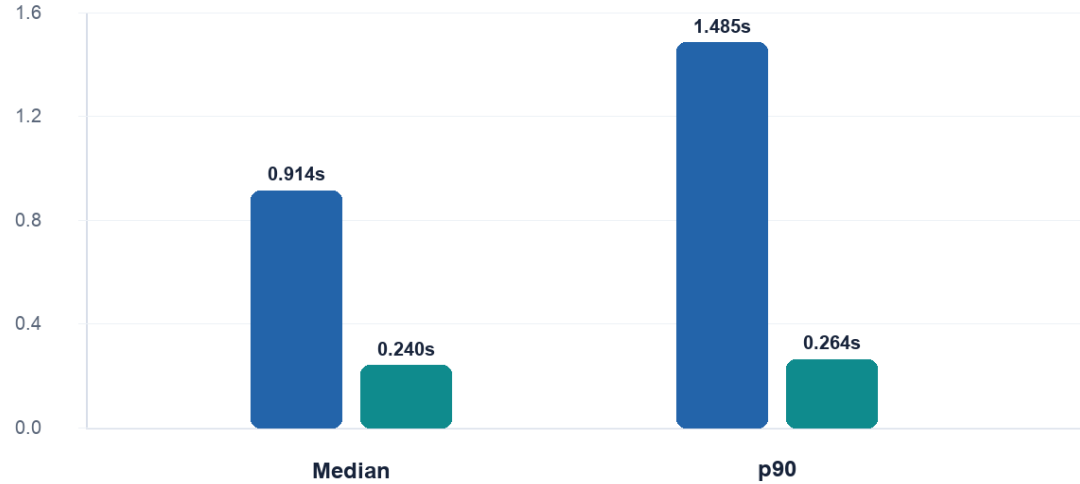
5.6x

p90 speedup
1.485s / 0.264s

Warm cleanup latency: MLX vs. Ollama

Lower is better. Values are measured over the same 24 reference transcripts.

Seconds



Executive Summary

The benchmark supports making MLX cleanup the center of the next Local Flow hardening phase. With Qwen2.5 1.5B Instruct 4-bit loaded through MLX, the cleanup hop dropped from a 0.914 second Ollama median to 0.240 seconds while keeping the dictation path local.

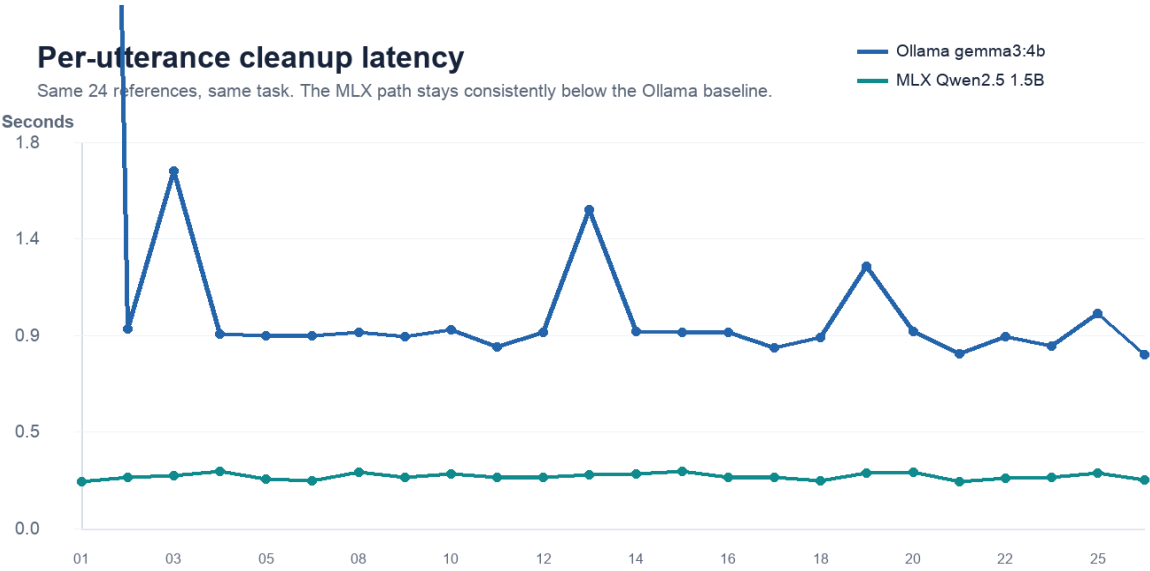
This is a performance proof, not a blind production swap. The MLX path still needs deterministic output sanitization and guardrails before becoming the default.

Research Method

Method item	Value
Input set	24 references from bench/samples/manifest.json
Baseline	Ollama gemma3:4b cleanup report
Candidate	mlx-community/Qwen2.5-1.5B-Instruct-4bit
Sampling	Temperature 0.0 strict rerun
Runtime condition	Qwen3-Coder 30B unloaded before final rerun
Decision standard	Lower latency only counts if meaning preservation remains enforceable

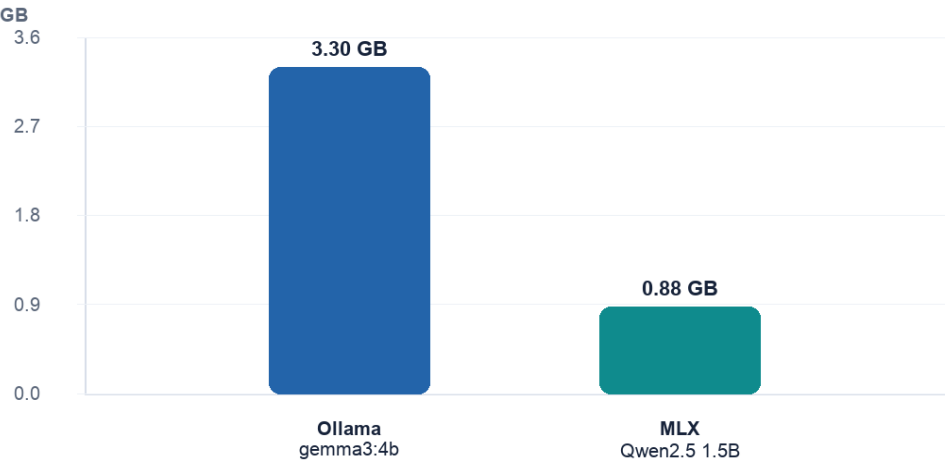
Measured Results

Dimension	Ollama gemma3:4b	MLX Qwen2.5 1.5B	Interpretation
Median warm cleanup	0.914s	0.240s	3.8x faster
p90 warm cleanup	1.485s	0.264s	5.6x faster at the tail
First-call/load effect	12.070s first call	0.669s load + warmup	Lower residency penalty
Model footprint	3.3 GB	0.88 GB cached	Materially lighter



Runtime model footprint

Smaller resident model improves the odds of fast cleanup under normal desktop load.



Architecture Implication

Cleanup architecture shift

Current baseline



Proposed hardening target



The product goal is fast local cleanup. MLX removes the Ollama daemon hop, uses a smaller model, and aligns the cleanup engine with Apple Silicon.

Quality Findings

Finding	Implication
Meaning preservation	No obvious answer-style hallucinations in the strict MLX rerun.
Known artifacts	Backticks around GROUP BY; one added article in 'the Standup'; some filler retained.
Production guard	Strip Markdown/code formatting, reject added content, preserve protected markers, fall back to raw.
Acceptance gate	24-reference acceptance set, production plausibility guard, and no non-local network dependency.

Recommendation

Scope Phase 3 Continuation as MLX Performance and Reliability Hardening: implement an optional MLX cleanup provider, deterministic sanitizer, benchmark reporting, and rollback to raw or Ollama when safety checks fail.

Acceptance target: MLX median below 0.350 seconds, p90 below 0.500 seconds, zero meaning changes, no Markdown artifacts, and no non-local network dependency in the dictation path.

Sources

bench/samples/manifest.json
raw-ollama-cleanup.md
raw-mlx-cleanup.md
app/Sources/LocalFlow/Cleaner.swift
<https://huggingface.co/mlx-community/Qwen2.5-1.5B-Instruct-4bit>
<https://huggingface.co/Qwen/Qwen2.5-1.5B-Instruct>